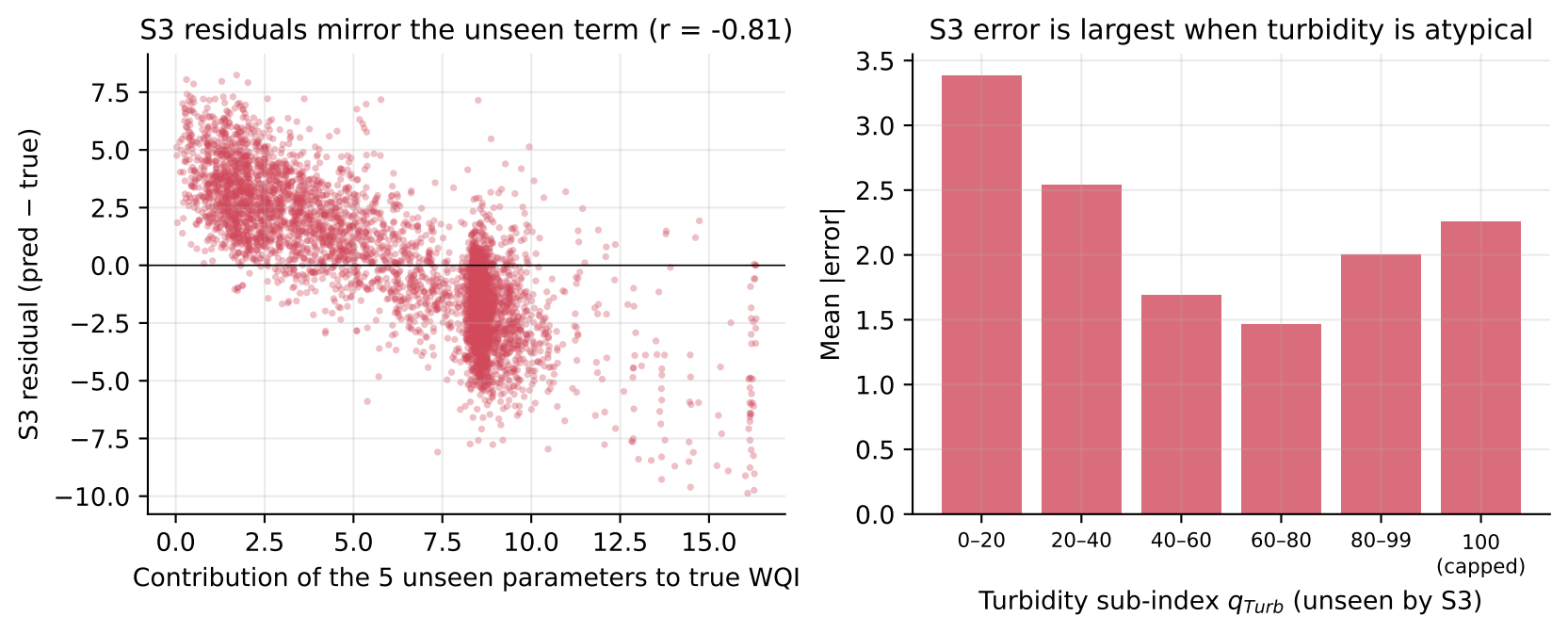

S3 residuals mirror the unseen term (r = -0.81)
S3 residual (pred − true)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Contribution of the 5 unseen parameters to true WQI
S3 error is largest when turbidity is atypical
Mean |error|
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0–20
20–40
40–60
60–80
80–99
100
(capped)
Turbidity sub-index q_{Turb} (unseen by S3)